

## ASSIGNMENT NO :04 (CLO3)

**Topic:** Transaction Processing and Concurrency Control

**Instructor:** Engr. Raheela Ambrin

**Group members:** Max 2

**Date of submission:** 19-05-2025

**Viva Date:** Asg 3 + 4 (Viva date:19-05-2025)

**Case Study:** ABC Markets sell products to customers. The relational diagram shown in Figure P10.6 represents the main entities for ABC's database. Note the following important characteristics:

- A customer may make many purchases, each one represented by an invoice. The CUS\_BALANCE is updated with each credit purchase or payment and represents the amount the customer owes.
  - The CUS\_BALANCE is increased (1) with every credit purchase and decreased (–) with every customer payment.
  - The date of last purchase is updated with each new purchase made by the customer.
  - The date of last payment is updated with each new payment made by the customer.
  - An invoice represents a product purchase by a customer.
  - An INVOICE can have many invoice LINEs, one for each product purchased.
  - The INV\_TOTAL represents the total cost of the invoice, including taxes.
  - The INV\_TERMS can be "30," "60," or "90" (representing the number of days of credit) or "CASH," "CHECK," or "CC."
  - The invoice status can be "OPEN," "PAID," or "CANCEL."
  - A product's quantity on hand (P\_QTYOH) is updated (decreased) with each product sale.
  - A customer may make many payments. The payment type (PMT\_TYPE) can be one of the following: "CASH" for cash payments. "CHECK" for check payments"" for credit card payments.
  - The payment details (PMT\_DETAILS) are used to record data about check or credit card payments:
    - The bank, account number, and check number for check payments.
    - The issuer, credit card number, and expiration date for credit card payments.
- Note:* Not all entities and attributes are represented in this example. Use only the attributes indicated.
- Using this database, write the SQL code to represent each of the following transactions. Use BEGIN TRANSACTION and COMMIT to group the SQL statements in logical

transactions.

a. On May 11, 2022, customer 10010 makes a credit purchase (30 days) of one unit of product 11QER/31 with a unit price of \$110.00; the tax rate is 8 percent. The invoice number is 10983, and this invoice has only one product line.

b. On June 3, 2022, customer 10010 makes a payment of \$100 in cash. The payment ID is 3428.

**Figure P10.6** The ABC Markets Relational Diagram

